ROBT613 · BRAIN–COMPUTER INTERFACES · MIDTERM PROJECT

BCI Inefficiency

Does resting or pre-trial alpha/mu power predicts which subjects decode well?

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01 · MOTIVATION

Not everyone can drive a motor-imagery BCI

15–30%

of users cannot reach reliable control with a motor-imagery BCI

Vidaurre & Blankertz, 2010

Calibration first, answer later

Each new user records a long calibration session before we learn whether decoding works for them.

We see it in our data too

In Cho2017, 19–20 of 52 subjects stay below AUC 0.6 with our decoder.

Our question

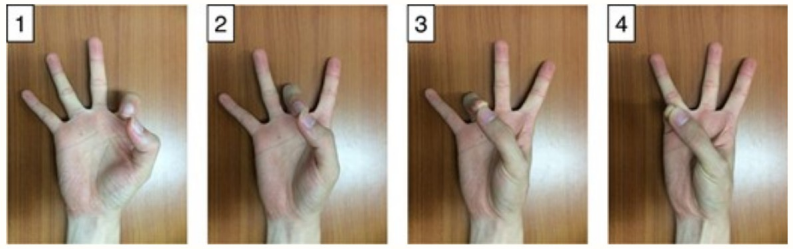
Can a short EEG measurement taken before the task predict who will decode well?

01 · MOTIVATION

Motor imagery: moving without moving

Motor imagery (MI): the user vividly imagines moving the left or right hand, without contracting any muscle.

The motor cortex still activates much as in real movement, and this change is visible in scalp EEG over C3 and C4.



1 · Imagine
Left or right hand, for 3 s after a cue

→

2 · Record
64-channel EEG over the sensorimotor cortex

→

3 · Decode
Band-power features into a linear classifier

→

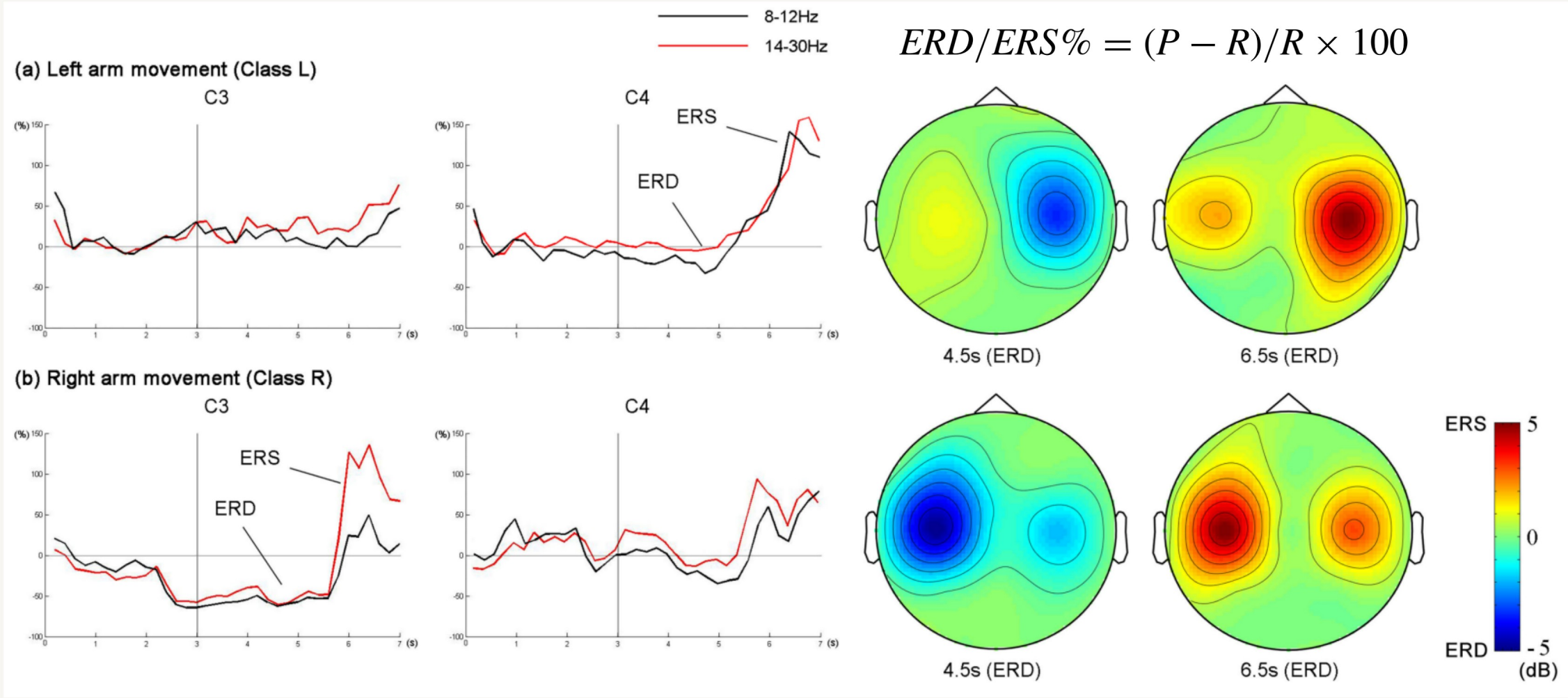
4 · Command
“Left” or “Right”, with no muscles involved

Brain rhythms: when do they appear?

Rhythm	Band	Where	When it appears / disappears
Alpha	8–13 Hz	Occipital (visual) cortex	Strong with eyes closed and relaxed; blocked when the eyes open or attention rises
Mu	8–13 Hz	Sensorimotor cortex (C3, Cz, C4)	Idle motor system; suppressed when we move, imagine a movement or watch one
Beta	13–30 Hz	Sensorimotor cortex	Holds the current motor state; drops during movement and rebounds above baseline right after it ends
Gamma	> 30 Hz	Local cortical patches	Active processing and movement onset; on the scalp mostly buried under muscle (EMG) noise

Our features use **8–30 Hz (mu + beta)**. Gamma is left out: in scalp EEG it mostly reflects muscle activity, not motor imagery.

ERD and ERS: relative power change

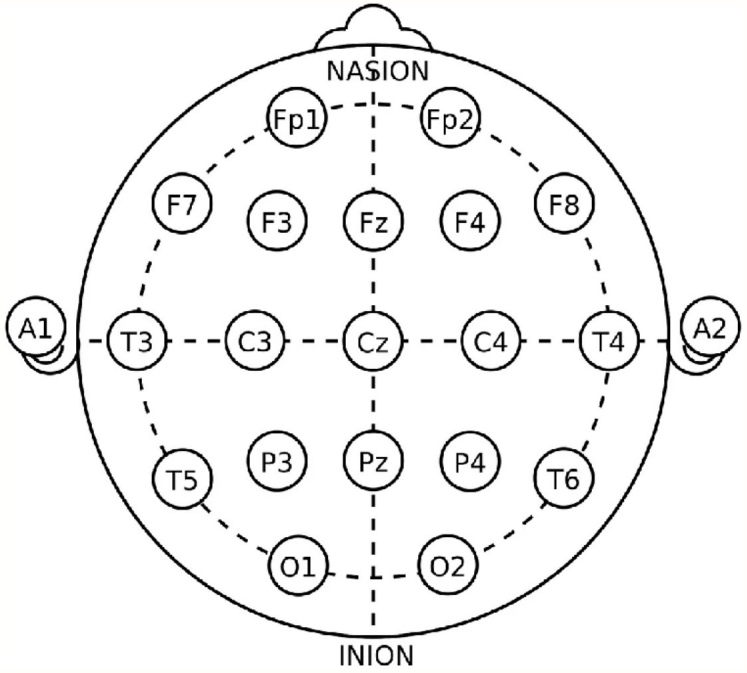


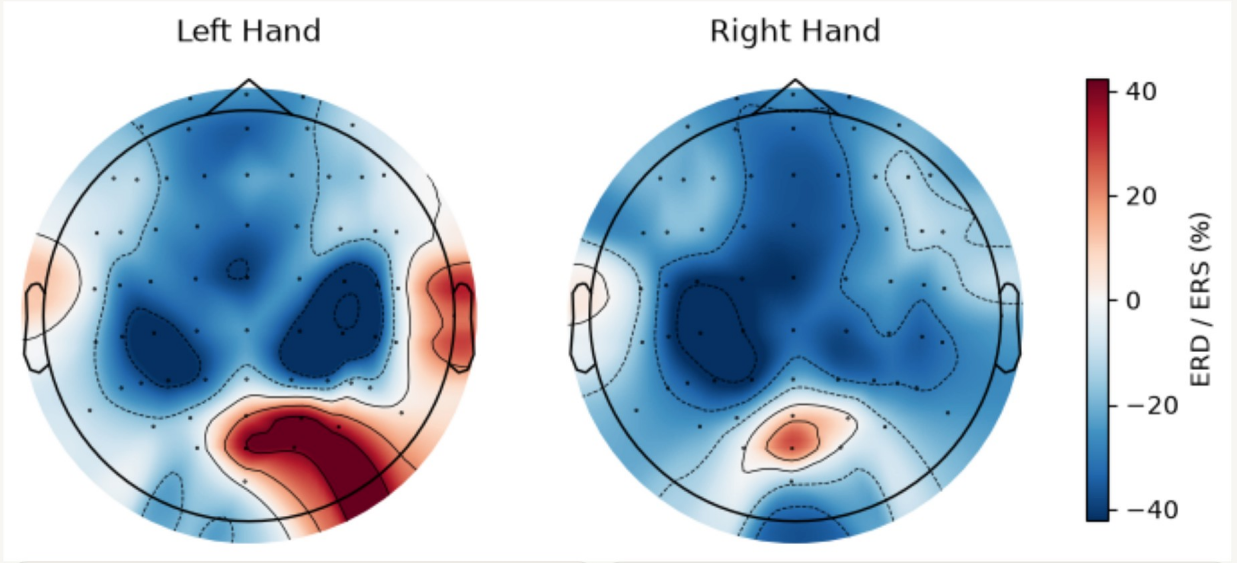
Tang, et al. 2019

ERD/ERS% < 0 - imagery
ERD/ERS% > 0 - back to rest

01 · PHYSIOLOGY

Each hemisphere controls the opposite hand





Imagine LEFT hand

→ ERD over **C4** (right hemisphere)

Imagine RIGHT hand

→ ERD over **C3** (left hemisphere)

01 · OUR HYPOTHESIS

“Subjects with stronger pre-trial mu power achieve better motor-imagery decoding.”

Strong idle mu
A large rhythm before the cue

→

Larger ERD
More rhythm to suppress during imagery

→

Higher AUC
Clearer left–right contrast for the classifier

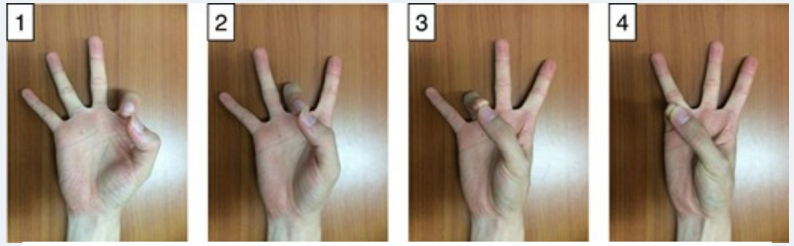
Test: Spearman correlation between each subject’s pre-trial log mu power and that subject’s held-out ROC AUC (N = 52).

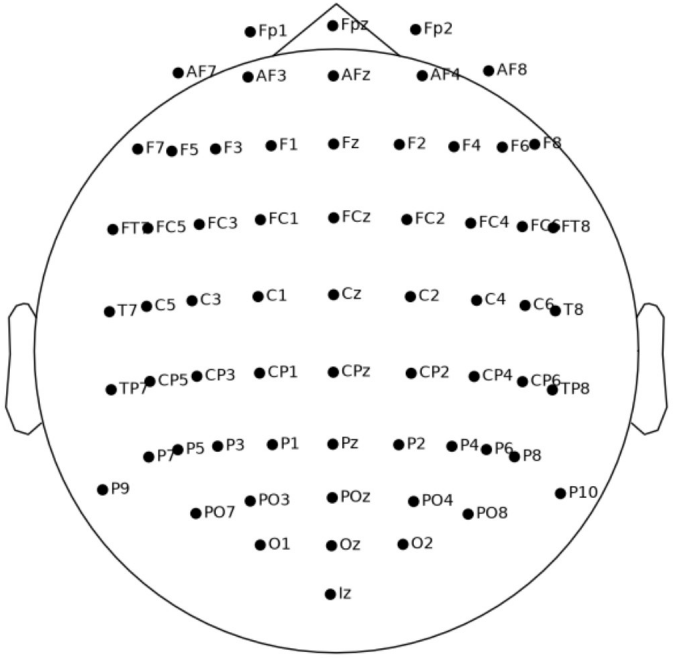
Blankertz et al., 2010 (resting SMR predictor, r = 0.53, N = 80) · Geronimo et al., 2016 (pre-trial mu) · Wang et al., 2022 (resting alpha, N = 105)

APPENDIX A1 · DATA AUDIT

Dataset (Cho et al., 2017)

Item	Value
Dataset	Cho2017 (GigaDB 100295), loaded with MOABB
Subjects	52 (IDs 1–52), all used; s20 and s33 ambidextrous
Sessions / runs	1 session; 5 runs (6 for s7, s9, s46), merged into one recording
Channels	64 EEG (10-10, Biosemi ActiveTwo) + 4 EMG + Stim
Sampling rate	512 Hz
Units	Volts
Reference	Not stated; we apply a common average reference
Events	1 = left hand, 2 = right hand (cue onset)
Trials	100 per class (120 for s7, s9, s46); 10,520 total
Epochs	−1.5 to 2.998 s, 2304 samples, no baseline correction





02 · DATA AUDIT

Cho2017: 52 people imagining hand movements

52
subjects (IDs 1–52), 1 session

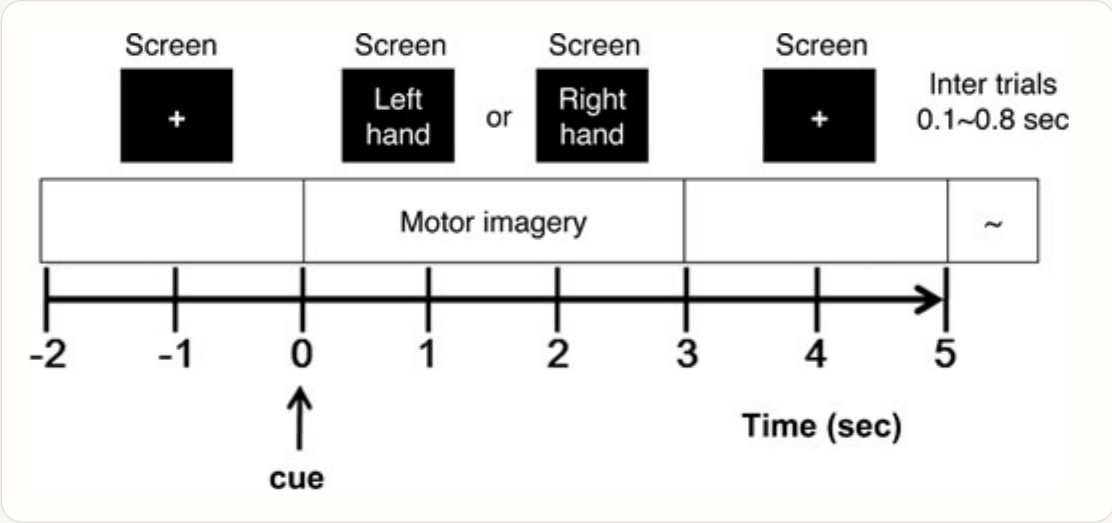
64 + 4
EEG (10-10) + EMG channels

512 Hz
sampling rate, units: volts

10,520
trials (200 per subject; 240 for s7, s9, s46)

1 / 2
event codes: left / right hand

CAR
reference not reported; we re-reference



Fixation cross for 2 s → cue “left hand” or “right hand” → 3 s of imagery → blank screen. 5 runs per subject, merged by MOABB into one recording.

02 · PREPROCESSING

Band-pass filter, then common average reference

1 · FIR band-pass 1–60 Hz

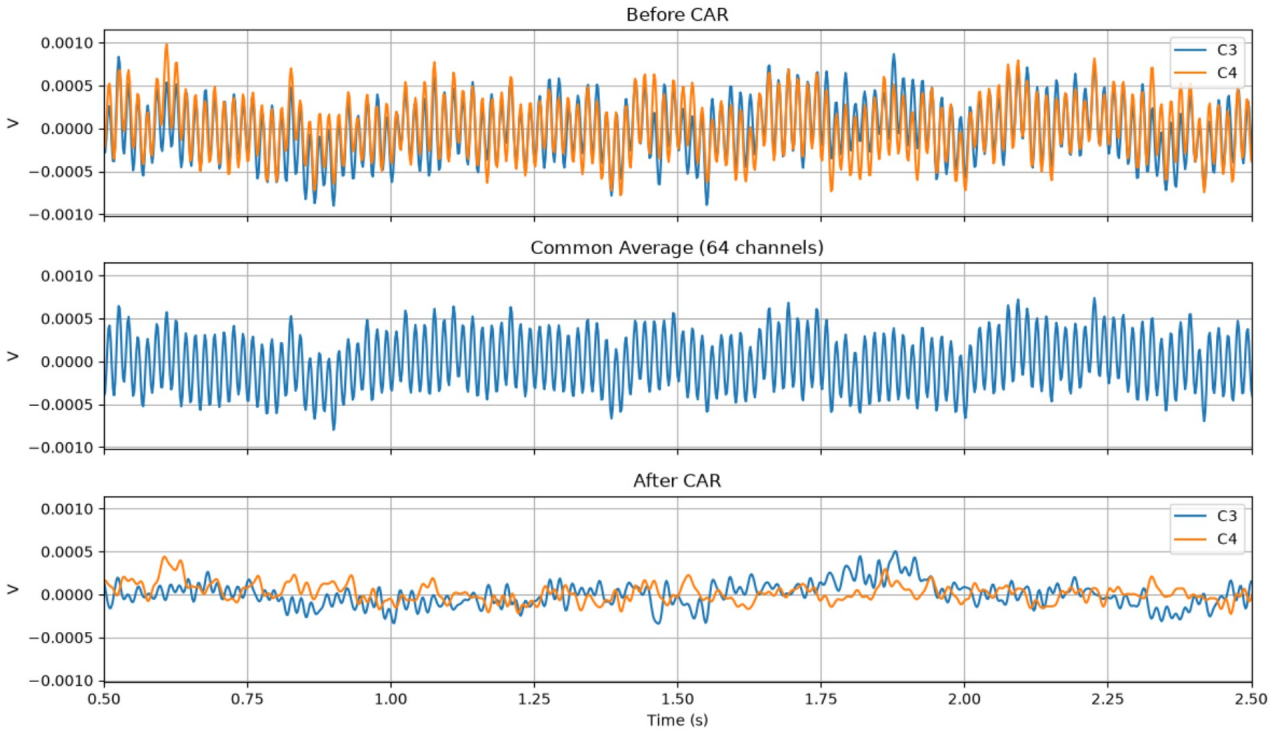
Hamming window, zero-phase. Removes the DC offset and slow drift; keeps mu and beta.

2 · Common average (CAR)

Subtract the mean of all 64 EEG channels at every sample. Removes activity shared by the whole cap.

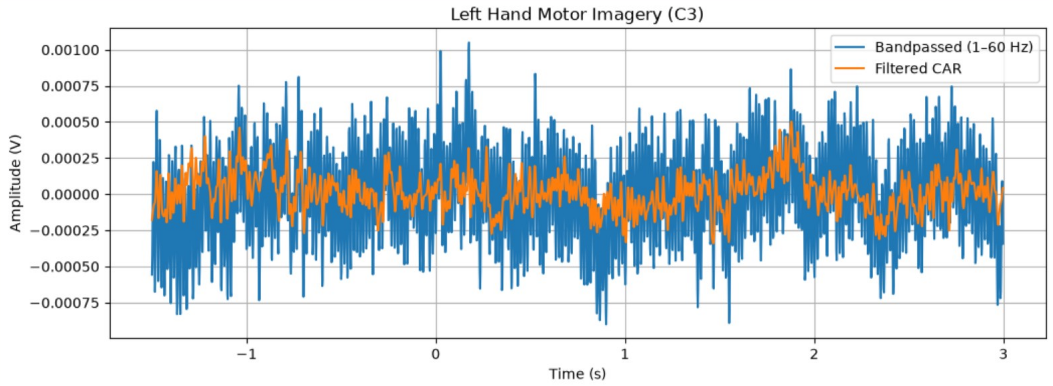
Effect of CAR

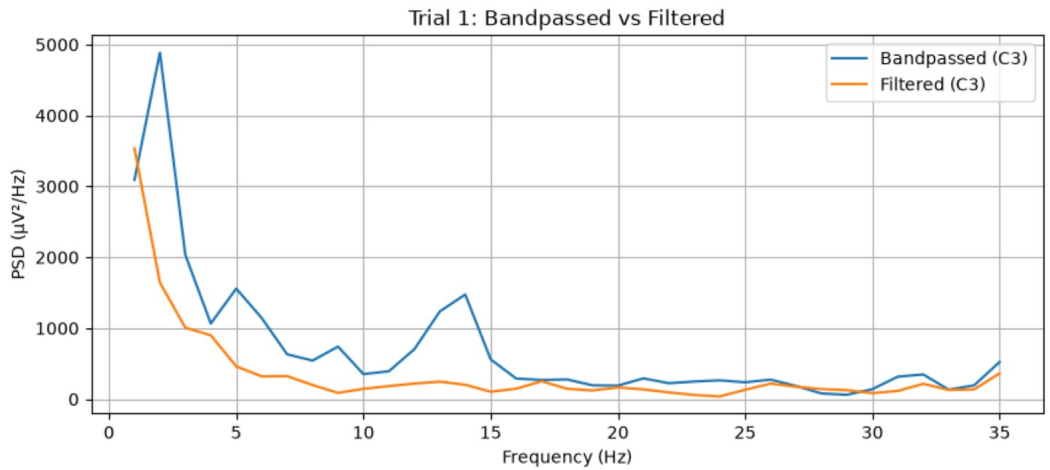
C3–C4 correlation: **0.845 → −0.140**
7 central channels: **0.888 → 0.374**



02 · PREPROCESSING

C3 before and after CAR: time and frequency

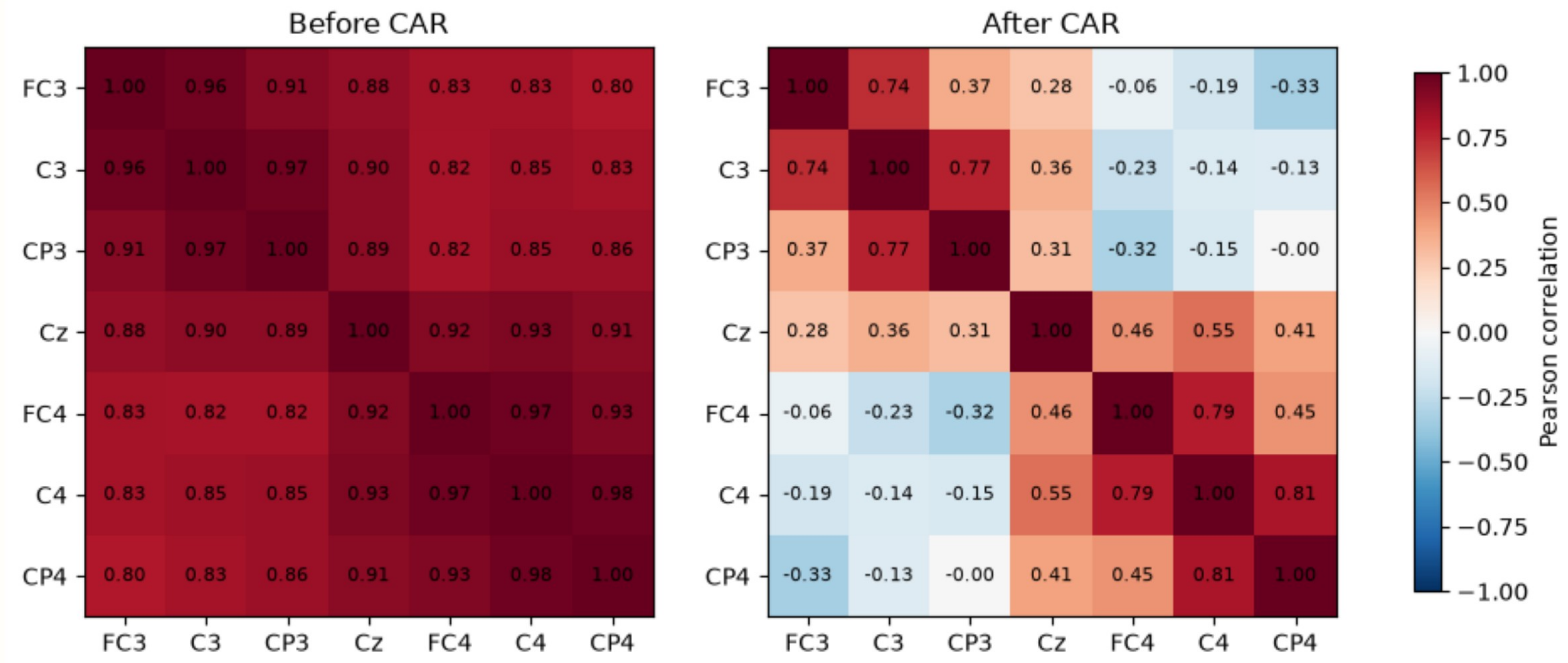




Both signals are already band-passed 1–60 Hz, so the difference is the effect of CAR alone. The 1 Hz high-pass removed the large constant offset of raw C3.

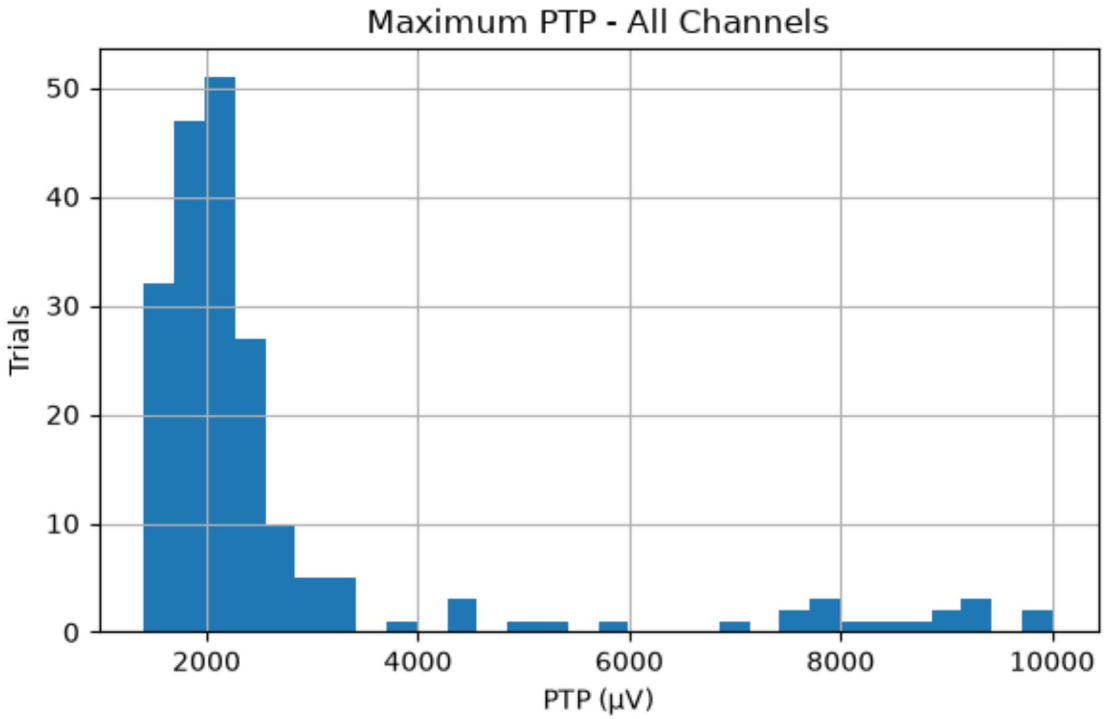
02 · PREPROCESSING

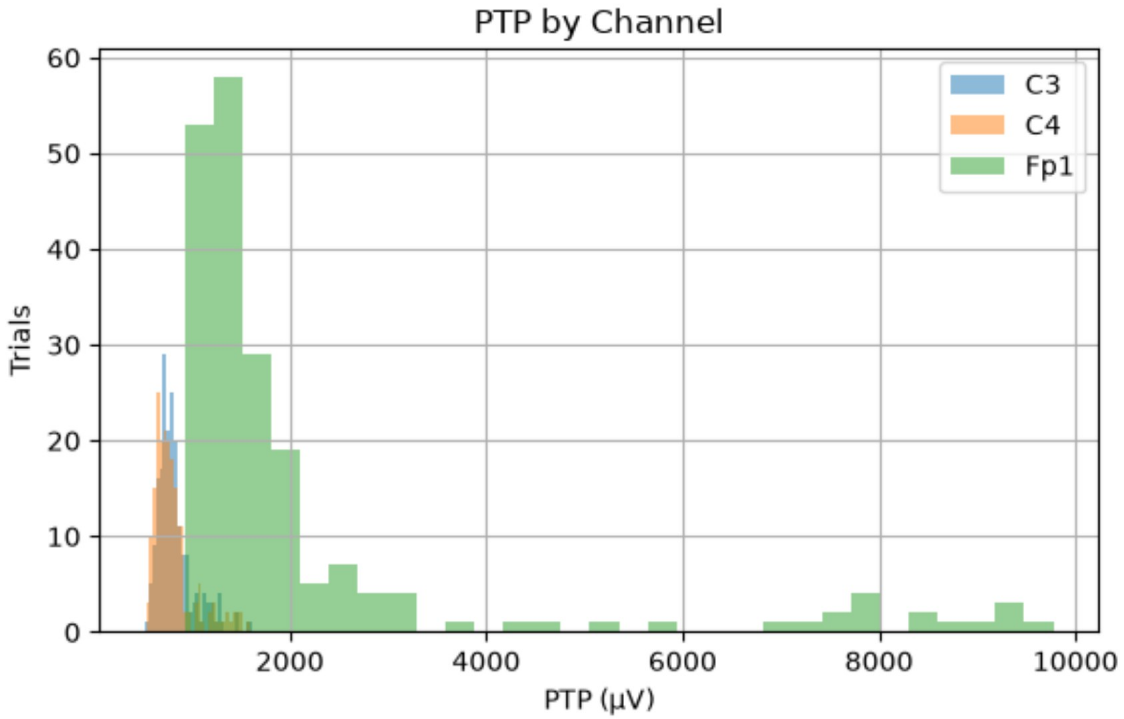
CAR lowers shared activity, not all correlation



Mean correlation of 7 central channels: 0.888 → 0.374 · C3–C4: 0.845 → −0.140 · Neighbours such as C3–FC3 stay correlated: they see the same cortex.

Large artifacts sit on frontal channels, not over C3/C4





Screening rule

Peak-to-peak per epoch, flagged if above median + 3 × MAD: 23 of 200 epochs.

Where

20 of 23 peak on Fp1/Fp2, 3 on AF7/AF8: likely eye blinks.

Decision: 0 rejected

Dropping hard test trials would inflate AUC; features use only central channels.

02 · ARTIFACT HANDLING

23 flagged epochs (subject 3), none rejected

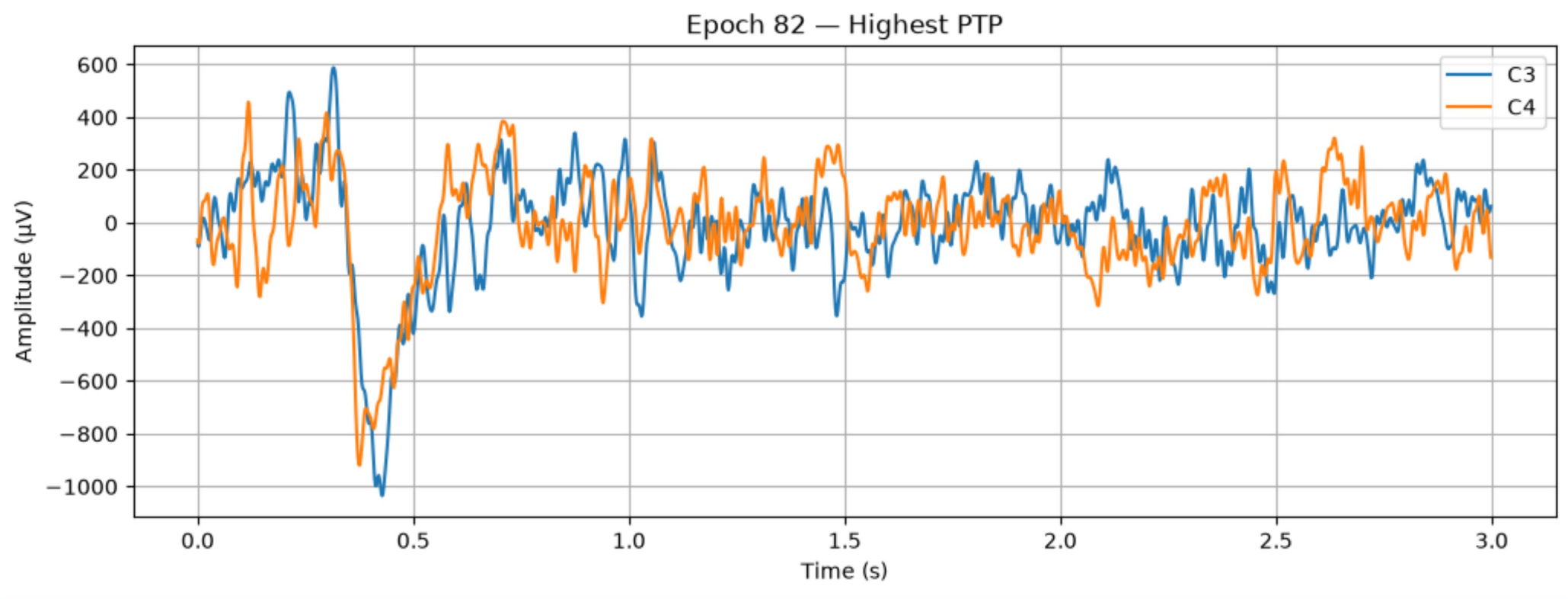
Epoch	Class	Worst	PTP µV
2	Left	Fp2	8552
10	Left	Fp1	9240
16	Left	AF7	3911
37	Left	Fp2	9848
39	Left	AF8	4356
58	Left	Fp2	8055
62	Left	Fp2	4466
74	Left	Fp1	7506

Epoch	Class	Worst	PTP µV
82	Left	Fp2	10002
89	Left	Fp1	5236
92	Left	Fp1	9376
96	Left	Fp1	7954
114	Right	Fp1	5899
117	Right	Fp1	4297
129	Right	Fp2	8874
141	Right	Fp2	9039

Epoch	Class	Worst	PTP µV
150	Right	Fp2	7988
158	Right	Fp2	7450
159	Right	AF7	4979
166	Right	Fp1	7724
175	Right	Fp2	9421
176	Right	Fp2	8670
190	Right	Fp1	7052

Rule: max PTP over 64 channels > median + 3 × MAD (median 2071 µV). 12 left, 11 right: balanced across classes.

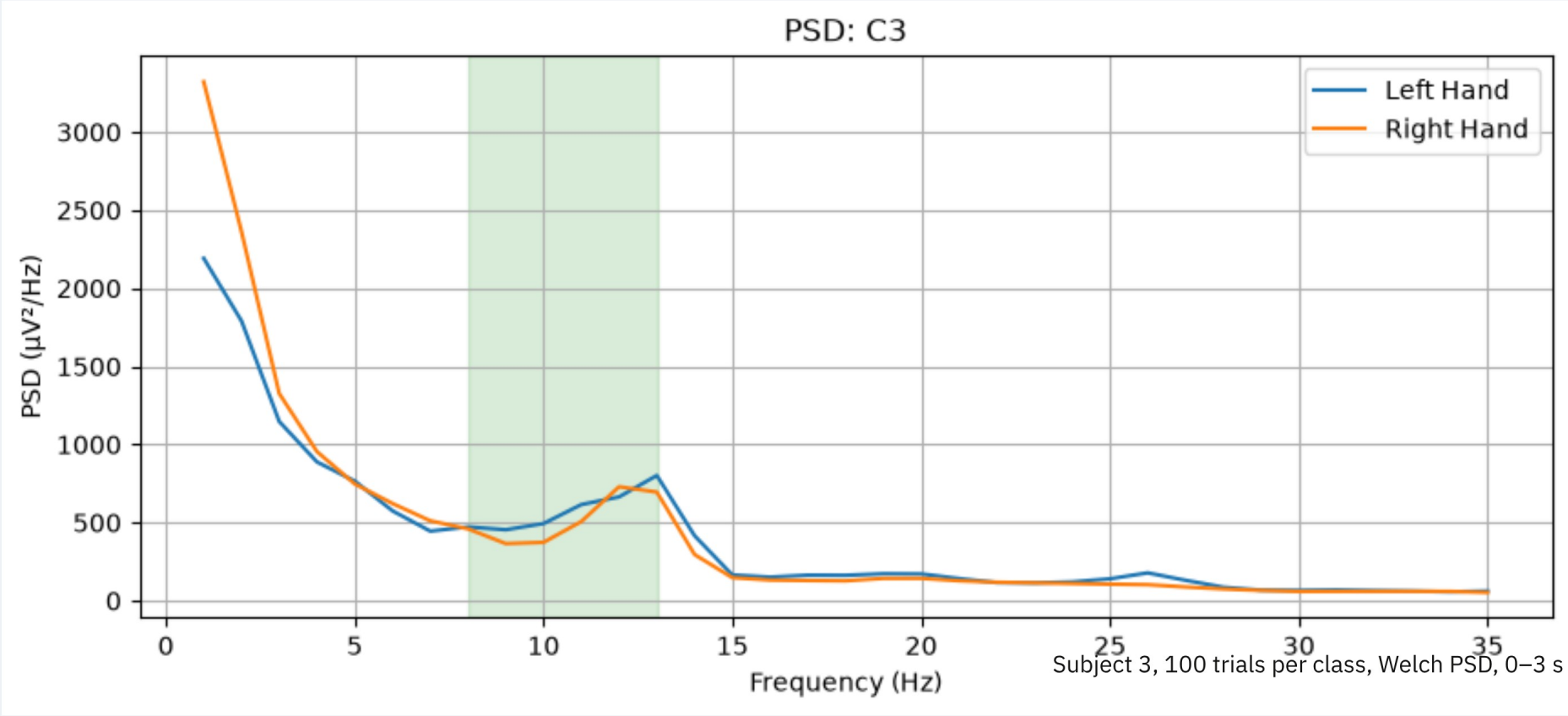
The worst C3/C4 epochs are the frontal ones



Top-5 PTP at C3: 82, 176, 2, 166, 117
Top-5 PTP at C4: 117, 92, 141, 2, 37

All 8 of these epochs are also on the frontal list (A4). A large frontal artifact enters the common average, so CAR spreads it to C3/C4.

Mu power is lower over the opposite hemisphere

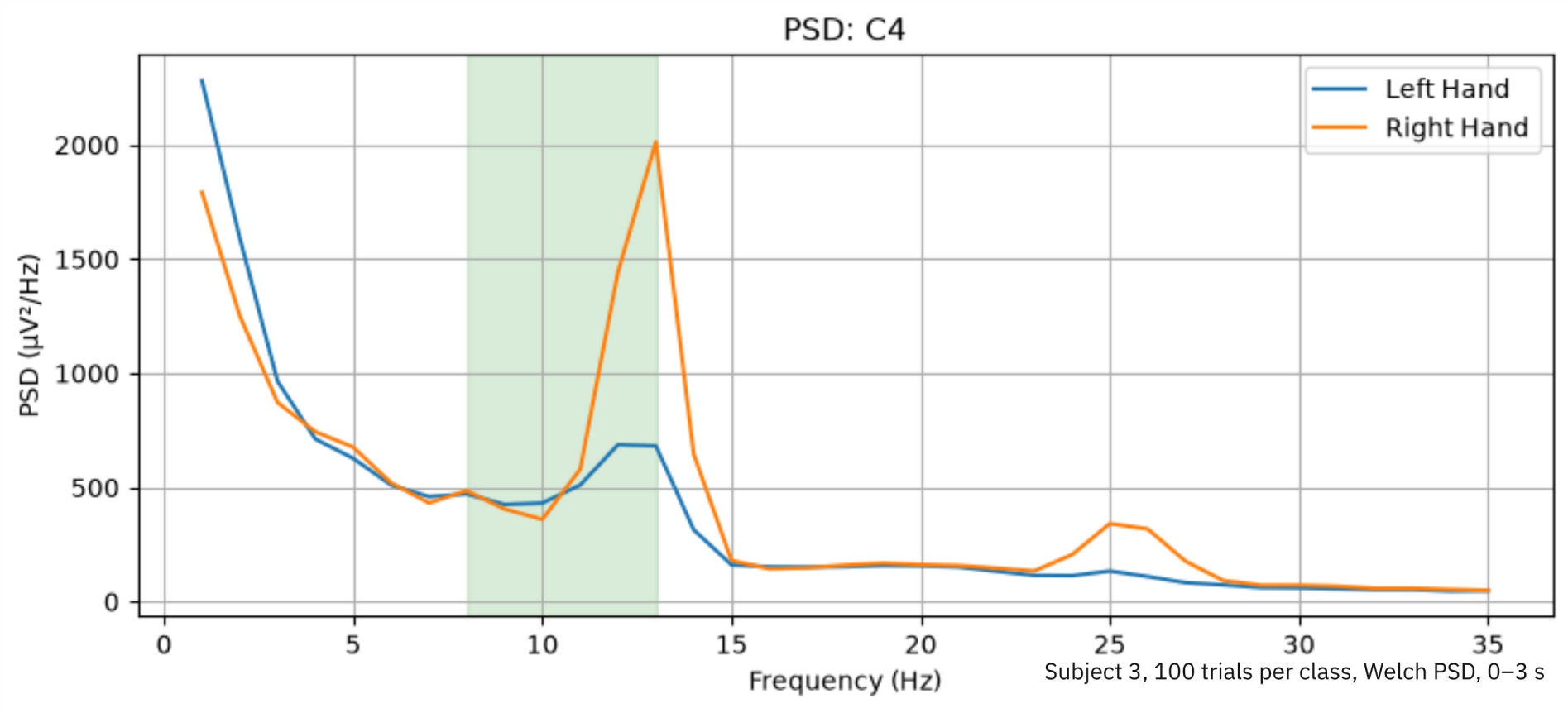


C3 (left hemisphere), mu 8–13 Hz

Left 2864 vs Right 2552 µV² → lower for the right hand, as expected

03 · NEUROPHYSIOLOGY CHECK

Mu power is lower over the opposite hemisphere



C4 (right hemisphere), mu 8–13 Hz

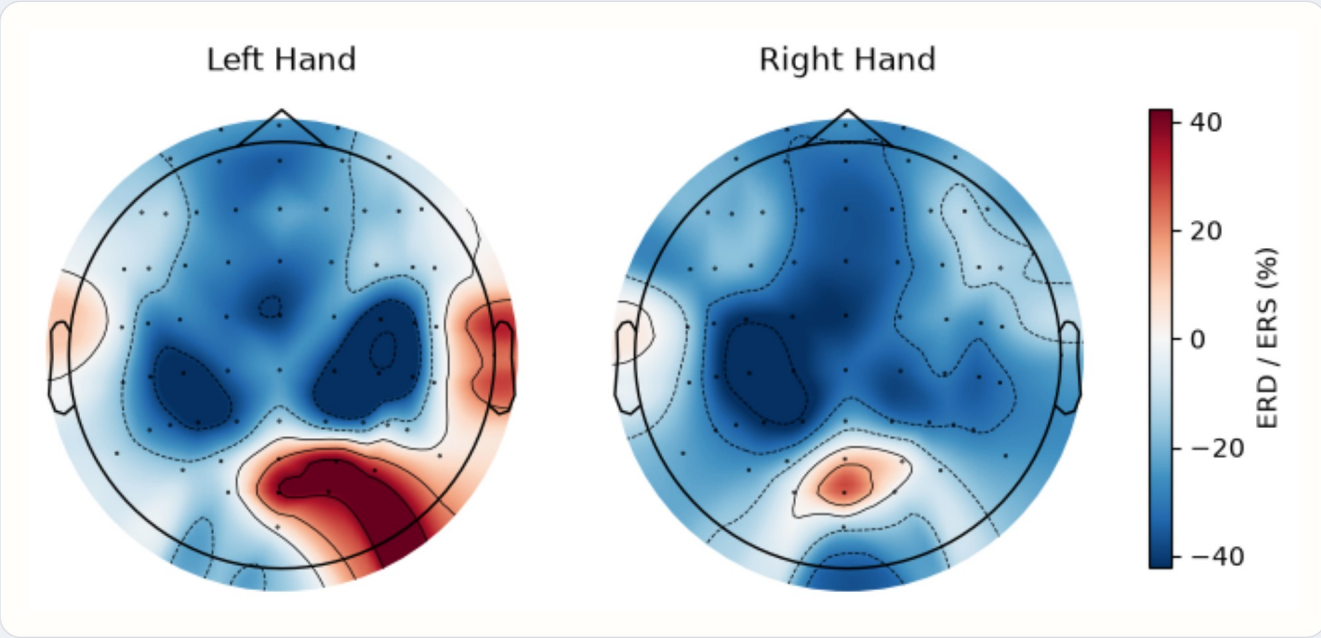
Left 2632 vs Right 4039 µV² → lower for the left hand, as expected

03 · NEUROPHYSIOLOGY CHECK

ERD is strongest over the contralateral side

8–30 Hz	C3	C4
Left hand	−28%	−56% ★
Right hand	−46% ★	−22%

★ = stronger ERD. Task 0.5–3 s vs pre-trial −1.5 to −0.5 s, median over trials. Both hands desynchronize both sides, but the opposite hemisphere drops about twice as much.



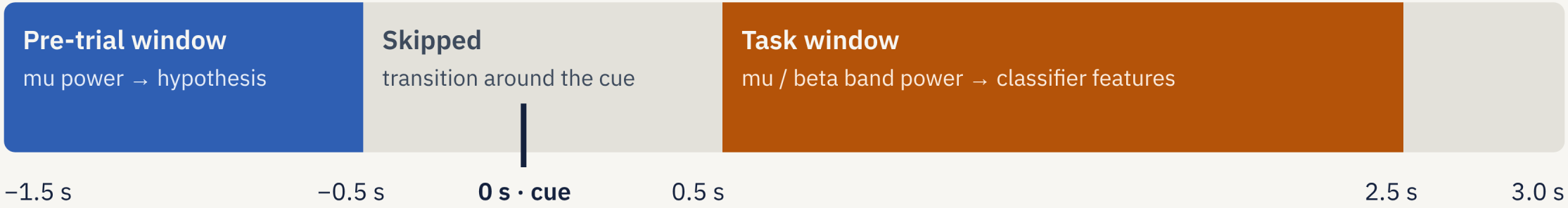
04 · FEATURES AND CLASSIFIER

52 log band-power features per trial



Two windows per trial that never overlap

“Subjects with stronger pre-trial mu power achieve better motor-imagery decoding.”



Epochs

−1.5 to 2.998 s around the cue: 2304 samples at 512 Hz, no baseline correction.

Predictor (per subject)

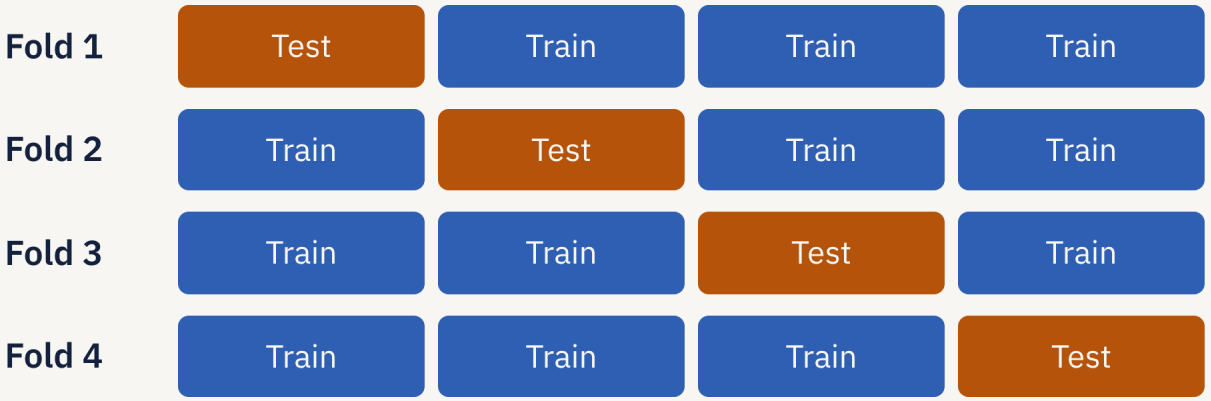
Log mu power, 8–13 Hz, −1.5 to −0.5 s, averaged over 13 channels and all trials.

Features (per trial)

Log band power, 8–30 Hz, 0.5 to 2.5 s, used to classify left vs right.

04 · HONEST EVALUATION

Test subjects are never seen in training



Each block = 13 subjects (all of their trials). GroupKFold(4), grouped by subject ID.

No leakage between people

~39 subjects train, ~13 test. Never a random split of epochs.

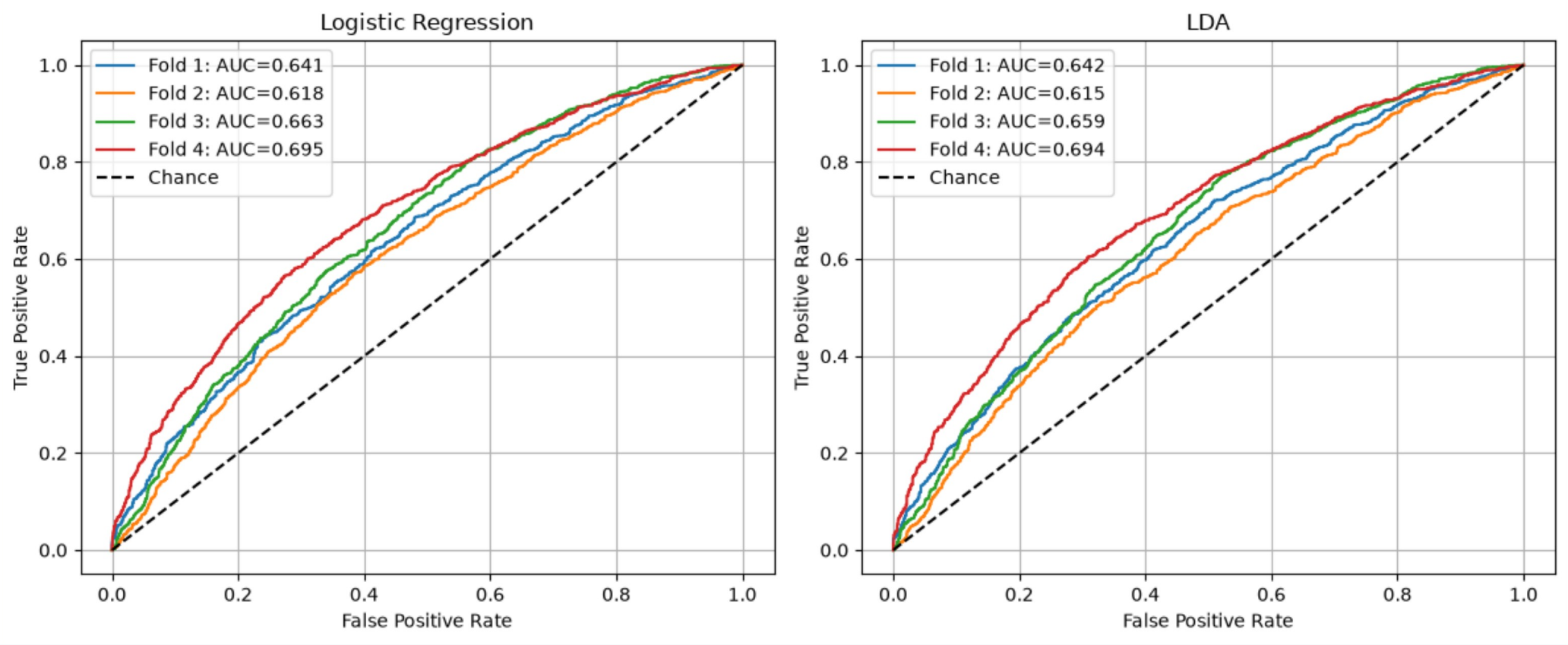
Nested tuning

Inner GroupKFold(3) on training subjects picks C from 0.001–10. C = 0.01 in all 4 folds.

Chance level

300 permutations of labels within subjects, same grouped CV.

ROC curves for each test fold

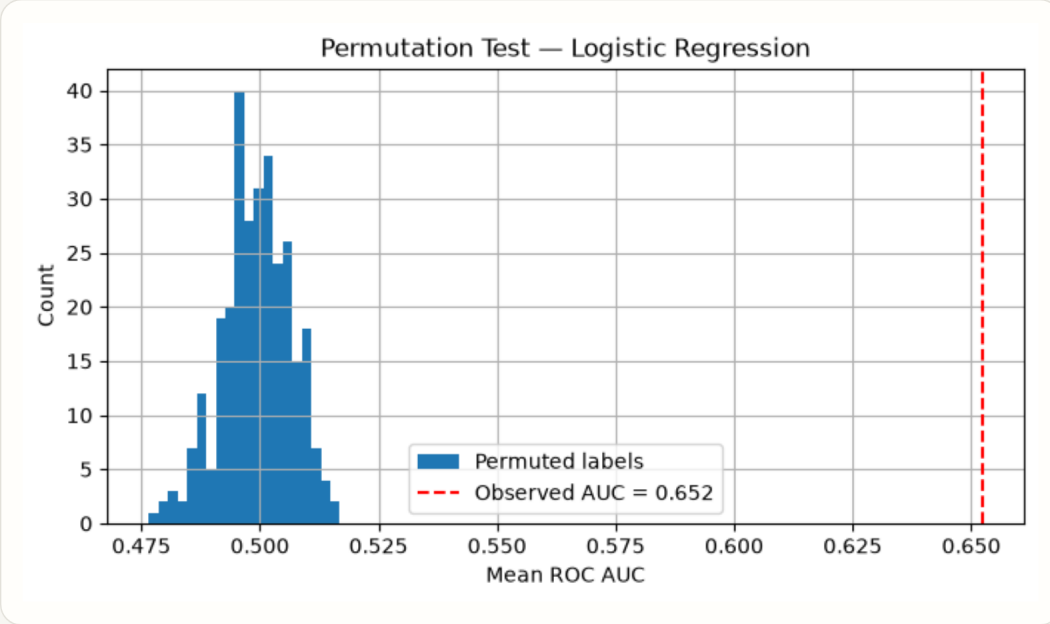


Fold AUCs · LogReg: 0.654+-0.033 · LDA: 0.652+-0.033

Cross-subject decoding is well above chance

Model	ROC AUC	Accuracy
Logistic regression	0.654 ± 0.033	0.609 ± 0.025
Shrinkage LDA	0.652 ± 0.033	0.608 ± 0.027
Chance (permuted)	0.499	0.5

Mean ± SD over 4 subject folds. Fold AUCs (LogReg): 0.641, 0.618, 0.663, 0.695. Classes are balanced, so accuracy equals balanced accuracy.



Permutation p = 0.0033

No shuffled run reached the real score: 1 / 301 is the smallest possible p with 300 permutations.

04 · Decoding results

ID	Mu	LR	LDA
1	9.48	0.700	0.702
2	7.90	0.553	0.560
3	8.31	0.952	0.947
4	9.09	0.784	0.775
5	9.41	0.652	0.639
6	7.73	0.584	0.583
7	7.64	0.610	0.607
8	7.79	0.543	0.536
9	8.31	0.754	0.744
10	8.63	0.739	0.749
11	9.10	0.557	0.562
12	9.59	0.706	0.708
13	8.01	0.557	0.542
14	9.70	0.985	0.984
15	7.81	0.754	0.755
16	8.42	0.699	0.685
17	7.07	0.430	0.426
18	7.48	0.696	0.699
19	7.99	0.670	0.664
20	8.89	0.666	0.667
21	8.66	0.576	0.582
22	8.34	0.740	0.752
23	8.69	0.762	0.745
24	7.61	0.787	0.782
25	7.99	0.745	0.751
26	8.52	0.558	0.555
27	7.26	0.499	0.491
28	7.78	0.454	0.453
29	6.89	0.566	0.569
30	7.18	0.701	0.694
31	8.44	0.691	0.652
32	7.77	0.539	0.539
33	8.68	0.616	0.615
34	7.59	0.522	0.524
35	7.38	0.483	0.493
36	8.37	0.694	0.676
37	7.58	0.513	0.511
38	7.62	0.539	0.527
39	9.50	0.637	0.637
40	8.28	0.446	0.461
41	9.20	0.929	0.927
42	7.39	0.714	0.699
43	8.19	0.977	0.971
44	8.90	0.675	0.651
45	8.96	0.655	0.662
46	8.84	0.663	0.673
47	9.12	0.722	0.718
48	9.66	0.887	0.882
49	8.45	0.599	0.604
50	8.40	0.565	0.562
51	8.18	0.595	0.594
52	8.13	0.736	0.732

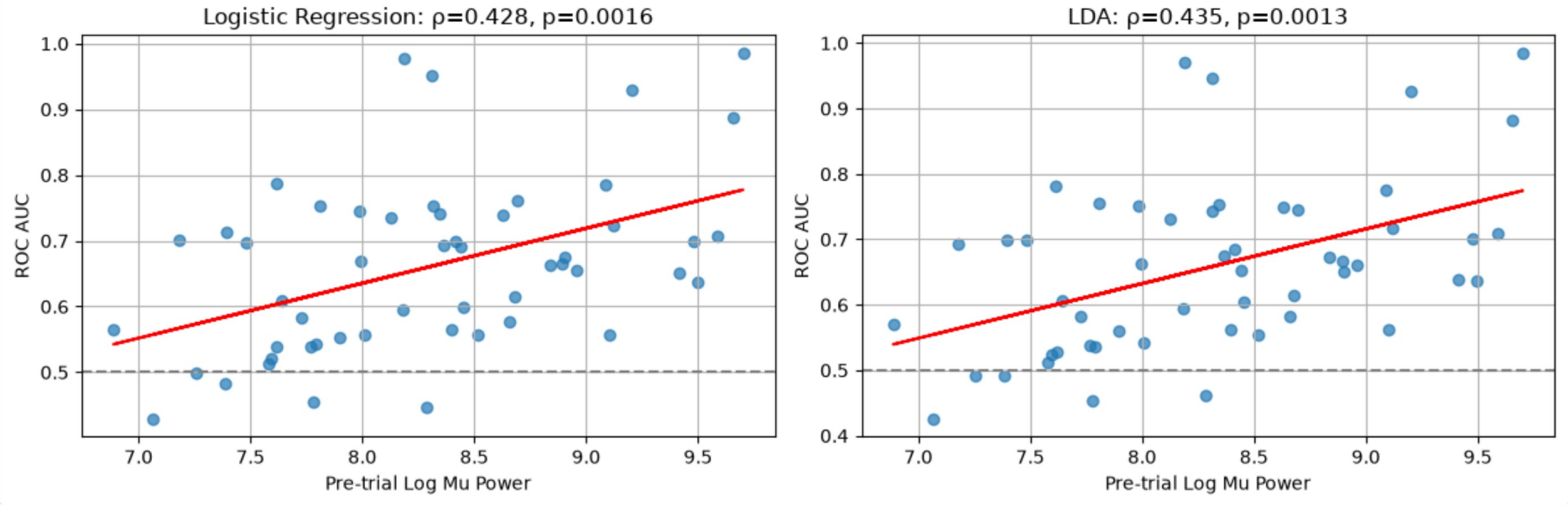
05 · HYPOTHESIS TEST

Higher pre-trial mu, better decoding

Logistic regression
ρ = 0.43 · p = 0.0016

Shrinkage LDA
ρ = 0.44 · p = 0.0013

Spearman, N = 52 subjects
Supported



BCI Inefficiency · ROBT613 Midterm · One dot = one subject · dashed line = AUC 0.5 · red = least-squares trend

05 · CLASSIFIER COMPARISON

LogReg vs LDA: significant, but practically identical

+0.003

mean AUC difference per subject, in favour of logistic regression

p = 0.043

Wilcoxon signed-rank test, paired over 52 subjects

0.43 / 0.44

Spearman ρ with pre-trial mu for LogReg / LDA: the conclusion does not depend on the model

A difference of 0.003 AUC is far below the fold-to-fold SD of 0.033. With 52 paired subjects, the test detects a consistent but negligible edge.

05 · CONCLUSION

What we found, and what it does not show

Found

- Contralateral mu/beta ERD is present (C4 −56% for left, C3 −46% for right)
- Cross-subject decoder beats chance: AUC 0.65, $p = 0.003$
- Pre-trial mu tracks per-subject AUC: $\rho \approx 0.43$, $p \approx 0.002$

Limits

- Correlation, not causation; $\rho^2 \approx 0.19$ leaves most variance unexplained
- Pre-trial window comes from the task session, not a separate rest recording
- Absolute log power across subjects: high-mu people may just fit the shared model better
- One dataset, offline, no epoch rejection

A short pre-trial mu measurement could flag likely low performers before a long calibration session.

APPENDIX

Versions, seeds and subjects

Package	Version
Python	3.11.2
MNE-Python	1.13.2
MOABB	1.7.2
scikit-learn	1.9.1
SciPy	1.17.1
NumPy	2.4.6
pandas	3.1.0
matplotlib	3.11.2

Subjects

All 52 Cho2017 subjects (IDs 1–52); subject 3 for single-subject figures.

Randomness

GroupKFold is deterministic; permutation test uses random_state = 42.

Pipeline

One notebook, top to bottom: load → filter → CAR → epoch → features → CV → statistics.

APPENDIX

Roles: everyone leads one area and reviews another

Alar Akilbekov

Data and decoding lead

Loading, audit, event coding; features, Pipeline, GroupKFold, nested CV

Reviews: evaluation · Presents: 1–3, 8–9, 14–15

Aruzhan Kaltay

Signal processing lead

Filtering, CAR, artifact screening, PSD and ERD check, topomaps

Reviews: data audit · Presents: 4–6, 10–13

Alim Daniyarov

Evaluation and reporting lead

Chance level, permutation, Spearman and Wilcoxon tests, report, slides

Reviews: decoding, signal processing · Presents: 7, 16–19

AI tools

ChatGPT (OpenAI): studying the libraries (MNE, MOABB, scikit-learn) and methods. Claude Code (Anthropic): slides and report drafted from notebook outputs, pipeline review, follow-up analyses in recommendations/. Every number comes from our executed notebooks.

APPENDIX

References

1. Blankertz B. et al. (2010). Neurophysiological predictor of SMR-based BCI performance. *NeuroImage*.
2. Wang et al. (2022). Resting-state EEG in alpha rhythm may be indicative of the performance of motor imagery-based brain–computer interface. *Entropy*.
3. Geronimo A., Kamrunnahar M., Schiff S. J. (2016). Single trial predictors for gating motor-imagery brain–computer interfaces based on sensorimotor rhythm and visual evoked potentials. *Frontiers in Neuroscience*.
4. Cho H., Ahn M., Ahn S., Kwon M., Jun S. C. (2017). EEG datasets for motor imagery brain–computer interface. *GigaScience*.
5. Pfurtscheller G., Lopes da Silva F. H. (1999). Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clinical Neurophysiology*.
6. Vidaurre C., Blankertz B. (2010). Towards a cure for BCI illiteracy. *Brain Topography*.
7. Gramfort A. et al. (2013). MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*.
8. Jayaram V., Barachant A. (2018). MOABB: trustworthy algorithm benchmarking for BCIs. *Journal of Neural Engineering*.

Questions?

Every team member can answer on any stage of the pipeline.

Code: github.com/alarxx/BCI_Midterm_Project

Colab: project_mini.ipynb (12 subjects) · project.ipynb (all 52)

Appendix: full results, per-subject table, reproducibility

FOLLOW-UP ANALYSES

Does the result survive stricter tests?

1 · Each subject gets their own decoder

2 · Features normalized per subject

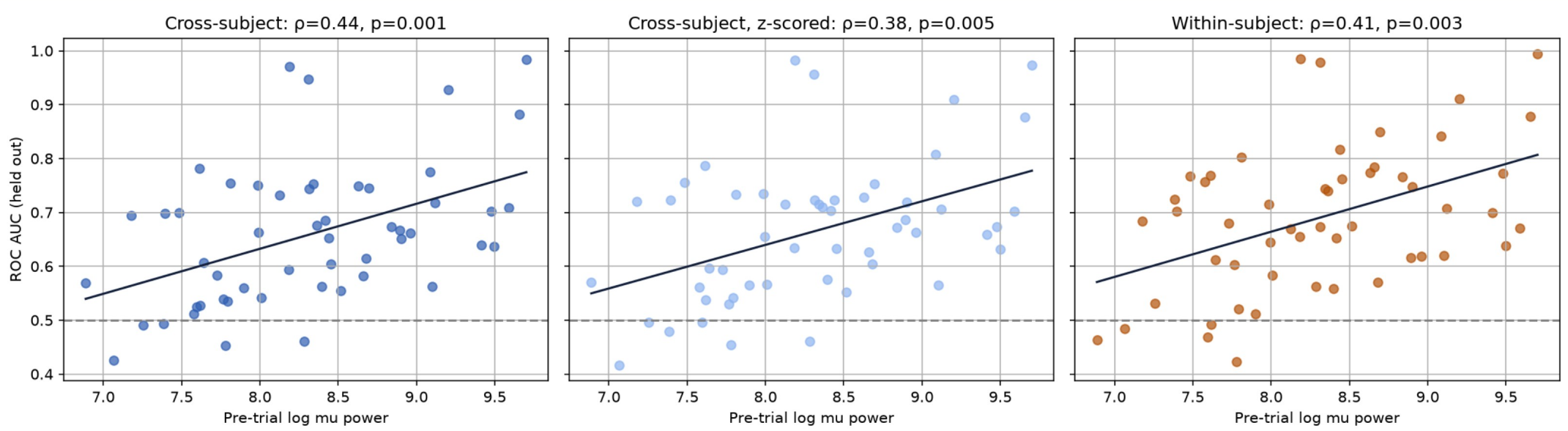
3 · Predictor from a separate rest recording

4 · Artifact rejection in training folds

Separate notebook: recommendations/recommendations.ipynb · project.ipynb unchanged · shrinkage LDA, same 52 features

FOLLOW-UP 1–2 · DECODER SETUP

The correlation is not an artefact of the shared model



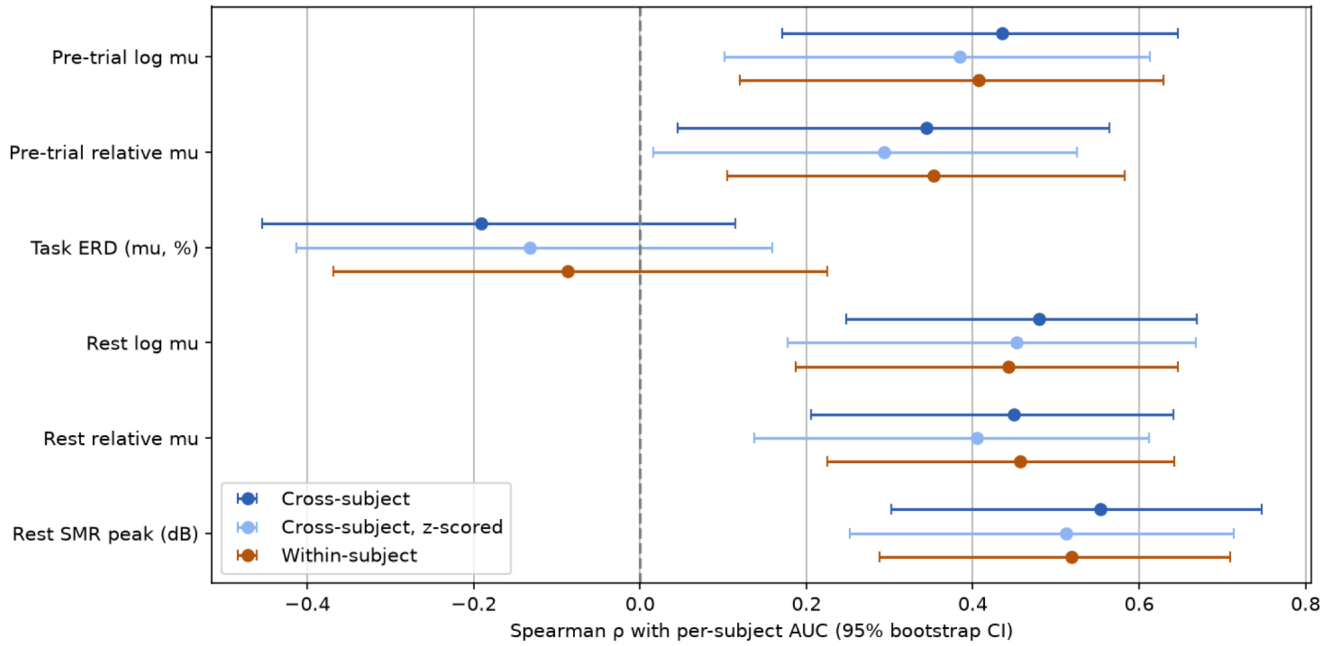
Cross-subject · ρ = 0.44
Main analysis, p = 0.001

Z-scored per subject · ρ = 0.38
Absolute power differences removed, p = 0.005

Within-subject · ρ = 0.41
Own LDA, leave-one-run-out, p = 0.003

FOLLOW-UP 3 · OTHER PREDICTORS

One minute of rest predicts even better



Rest SMR peak above 1/f

ρ = 0.55

$p = 2 \times 10^{-5}$ · Blankertz 2010: r = 0.53

Pre-trial vs rest mu: ρ = 0.82 across subjects; they measure the same trait.

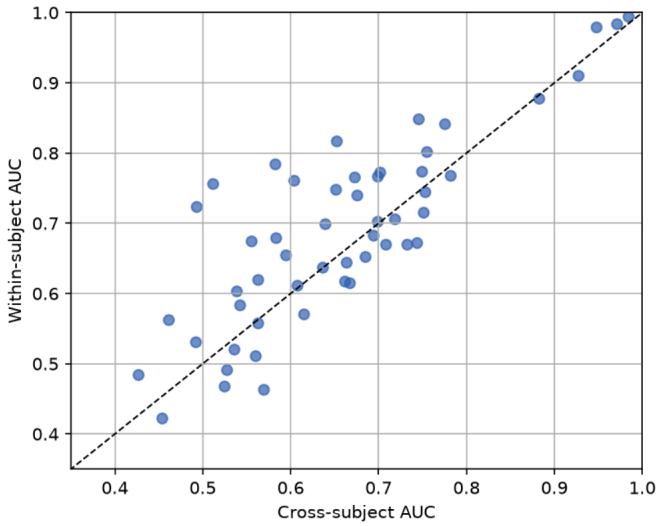
Relative mu (÷ 4–35 Hz) still works, ρ = 0.34–0.46: not only overall amplitude.

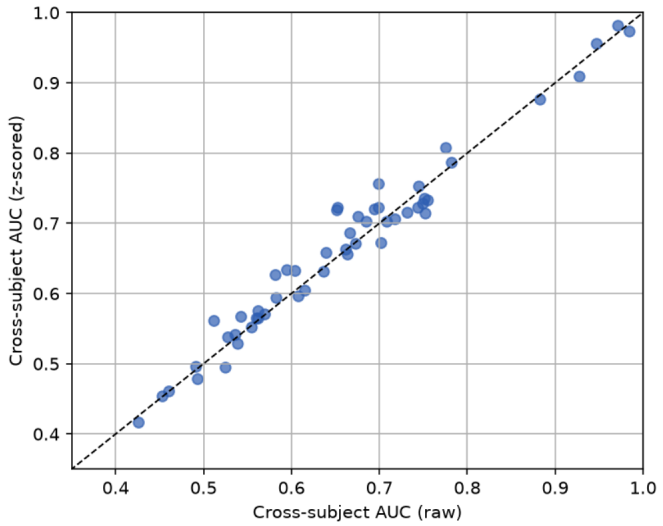
Task ERD (median, 13 ch., both hands): no correlation.

BCI Inefficiency · Follow-up · Rest: ~66 s recording in the original Cho2017 .mat files, not loaded by MOABB · bars = 95% bootstrap CI

FOLLOW-UP 1, 2, 4 · DECODING PERFORMANCE

Own decoders help; normalization and rejection do not





Decoder	AUC	< 0.6
Cross-subject	0.658	19
+ z-scoring	0.664	18
+ train rejection	0.658	19
Within-subject	0.690	12

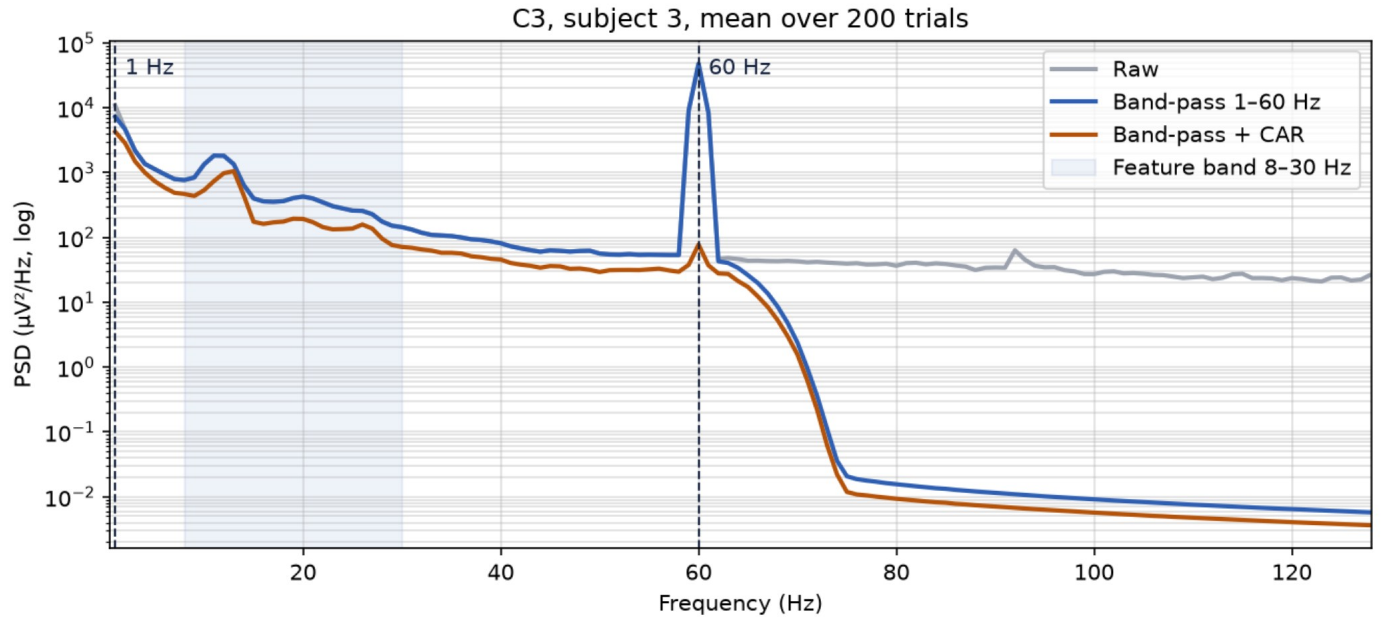
Within vs cross: +0.032, Wilcoxon p = 0.016. Same people are good in both (ρ = 0.75).

Z-scoring: +0.006, p = 0.19, not significant.

Rejection: 860 trials (8.2%) dropped from training: +0.0002, p = 0.59.

FOLLOW-UP 5 · SIGNAL INSPECTION

PSD before and after filtering: a 60 Hz surprise



Band-pass works

Above ~70 Hz power drops by more than 1000×; 8–30 Hz is unchanged.

60 Hz mains survives

Recorded in Korea (60 Hz grid); our upper edge is exactly 60 Hz.

CAR removes it

Line noise is common to all channels: >100× smaller after CAR. Better: a 60 Hz notch or a 40 Hz low-pass.